Matrices, Vectors & Complex Numbers

ANSWER KEY

Matrix arithmetic

1 Matrix A has rows $(2, 3)$ and $(-1, 4)$; matrix B has rows $(5, -2)$ and $(1, 3)$. Find the rows of $A + B$.

Adding corresponding entries: first row $(2+5,\ 3+(-2)) = (7, 1)$; second row $(-1+1,\ 4+3) = (0, 7)$.

2 Find $3A$ for matrix A with rows $(1, -2)$ and $(4, 0)$.

Multiply every entry by 3: rows $(3, -6)$ and $(12, 0)$.

Matrix multiplication

3 Matrix A has rows $(1, 2)$ and $(3, 4)$; matrix B has rows $(5, 6)$ and $(7, 8)$. Find AB.

Row 1 of AB: $(1(5)+2(7),\ 1(6)+2(8)) = (19, 22)$. Row 2 of AB: $(3(5)+4(7),\ 3(6)+4(8)) = (43, 50)$.

Determinants and inverses of 2×2 matrices

4 Find the determinant of the matrix with rows $(4, 2)$ and $(3, 5)$.

$\det = 4(5) - 2(3) = 20 - 6 = 14$.

5 Find the inverse of the matrix M with rows $(2, 1)$ and $(5, 3)$.

$\det(M) = 2(3) - 1(5) = 1$. $M^{-1} = \frac{1}{1} \times$ the matrix with rows $(3, -1)$ and $(-5, 2)$, giving M^{-1} with rows $(3, -1)$ and $(-5, 2)$.

Vectors: magnitude and direction

6 Find the magnitude of the vector $\mathbf{v} = (3, 4)$.

$|\mathbf{v}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

7 Given $\mathbf{a} = (2, -5)$ and $\mathbf{b} = (-3, 7)$, find $\mathbf{a} + \mathbf{b}$ and $2\mathbf{a} - \mathbf{b}$.

$\mathbf{a} + \mathbf{b} = (2-3,\ -5+7) = (-1, 2)$. $2\mathbf{a} - \mathbf{b} = (4-(-3),\ -10-7) = (7, -17)$.

The dot product

8 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 2)$ and $\mathbf{b} = (-1, 4)$, and determine whether the vectors are perpendicular.

$\mathbf{a} \cdot \mathbf{b} = 3(-1) + 2(4) = -3 + 8 = 5$. Since $\mathbf{a} \cdot \mathbf{b} \neq 0$, the vectors are not perpendicular.

Complex number arithmetic

9 Simplify $(4+3i)+(2-5i)$.
$(4+2)+(3-5)i=6-2i$.

10 Simplify $(3+2i)(1-4i)$.
$(3+2i)(1-4i)=3-12i+2i-8i^2=3-10i+8=11-10i$ (using $i^2=-1$).

Complex conjugates and division

11 Find the complex conjugate of $z=6-5i$, and compute $z\bar{z}$.
$\bar{z}=6+5i$. $z\bar{z}=(6-5i)(6+5i)=36-25i^2=36+25=61$.

12 Simplify $\frac{5+i}{2-i}$ by multiplying by the conjugate of the denominator.
$\frac{5+i}{2-i}\cdot\frac{2+i}{2+i}=\frac{(5+i)(2+i)}{4+1}=\frac{10+5i+2i+i^2}{5}=\frac{10+7i-1}{5}=\frac{9+7i}{5}=\frac{9}{5}+\frac{7}{5}i$.

Solving quadratics with complex solutions

13 Solve $x^2+2x+5=0$, giving your answer in the form $a+bi$.
Using the quadratic formula: $x=\frac{-2\pm\sqrt{4-20}}{2}=\frac{-2\pm\sqrt{-16}}{2}=\frac{-2\pm4i}{2}=-1\pm2i$.

Combining matrices and systems of equations

14 This question has two parts. The system $3x+2y=16$ and $x+4y=18$ can be written as a matrix equation $M\mathbf{v}=\mathbf{c}$, where M has rows $(3,2)$ and $(1,4)$. (a) Find $\det(M)$ and M^{-1}. (b) Use M^{-1} to solve the system for x and y.
(a) $\det(M)=3(4)-2(1)=10$. $M^{-1}=\frac{1}{10}\times$ the matrix with rows $(4,-2)$ and $(-1,3)$, giving rows $(0.4,-0.2)$ and $(-0.1,0.3)$. (b) Applying M^{-1} to $(16,18)^T$: $x=0.4(16)-0.2(18)=6.4-3.6=2.8$. $y=-0.1(16)+0.3(18)=-1.6+5.4=3.8$.

Powers of i and simplifying complex expressions

15 Simplify i^{17}, using the fact that powers of i cycle with period 4.
$17=4(4)+1$, so $i^{17}=(i^4)^4\cdot i^1=1^4\cdot i=i$.

16 Simplify $(2i)^3$.
$(2i)^3=8i^3=8(i^2\cdot i)=8(-1\cdot i)=-8i$.

Vector direction and unit vectors

17 Find a unit vector in the same direction as $\mathbf{v} = (6, -8)$.

$|\mathbf{v}| = \sqrt{36 + 64} = \sqrt{100} = 10$. Unit vector: $\frac{1}{10}(6, -8) = (0.6, -0.8)$.

18 Find the angle a vector $\mathbf{v} = (3, 3)$ makes with the positive x-axis.

$\theta = \arctan\left(\frac{3}{3}\right) = \arctan(1) = 45^\circ$.

Identity matrix and matrix equations

19 Explain why $MI = M$ for any matrix M and identity matrix I of matching size, and verify this for M with rows $(2, 3)$ and $(1, 4)$ multiplied by the 2×2 identity matrix (rows $(1, 0)$ and $(0, 1)$).

The identity matrix leaves any matrix unchanged under multiplication, just as the number 1 does in ordinary multiplication. Verifying: row 1 of MI: $(2(1) + 3(0),\ 2(0) + 3(1)) = (2, 3)$; row 2: $(1(1) + 4(0),\ 1(0) + 4(1)) = (1, 4)$ — identical to M's original rows, confirming $MI = M$.

A multi-part complex number problem

20 This question has two parts. Given $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. (a) Find $|z_1|$ (the modulus of z_1). (b) Find $z_1 \cdot z_2$ and its modulus, then verify that $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$.

(a) $|z_1| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$. (b) $z_1 z_2 = (3 + 4i)(1 - 2i) = 3 - 6i + 4i - 8i^2 = 3 - 2i + 8 = 11 - 2i$. $|z_1 z_2| = \sqrt{11^2 + 2^2} = \sqrt{121 + 4} = \sqrt{125} = 5\sqrt{5}$. $|z_2| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$. Check: $|z_1| \cdot |z_2| = 5\sqrt{5}$, matching $|z_1 z_2| = 5\sqrt{5}$ ✓.

Scalar multiplication of matrices and vectors combined

21 Given vectors $\mathbf{u} = (4, -3)$ and $\mathbf{v} = (-2, 5)$, find $3\mathbf{u} - 2\mathbf{v}$.

$3\mathbf{u} = (12, -9)$. $2\mathbf{v} = (-4, 10)$. $3\mathbf{u} - 2\mathbf{v} = (12 - (-4), -9 - 10) = (16, -19)$.